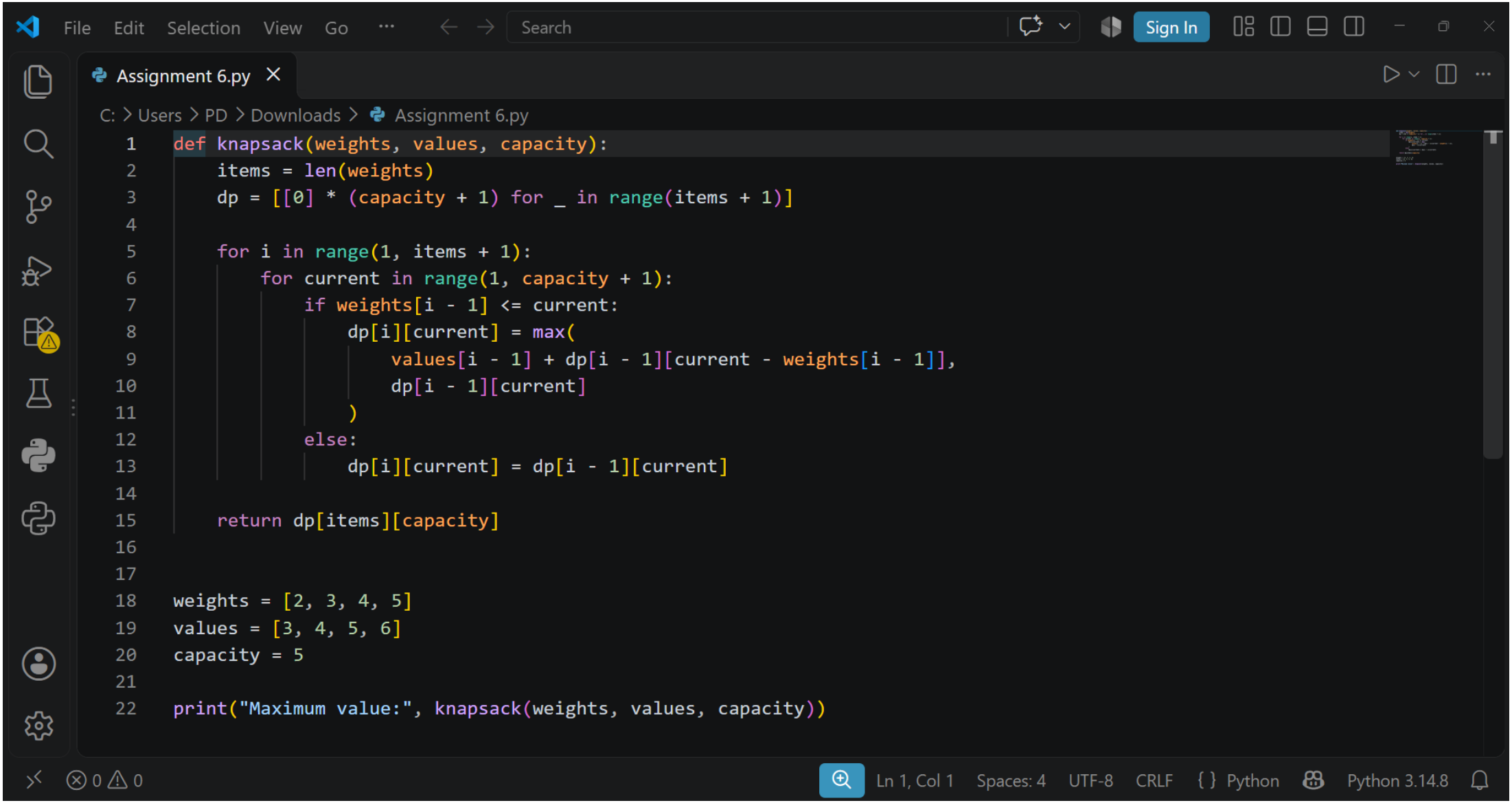
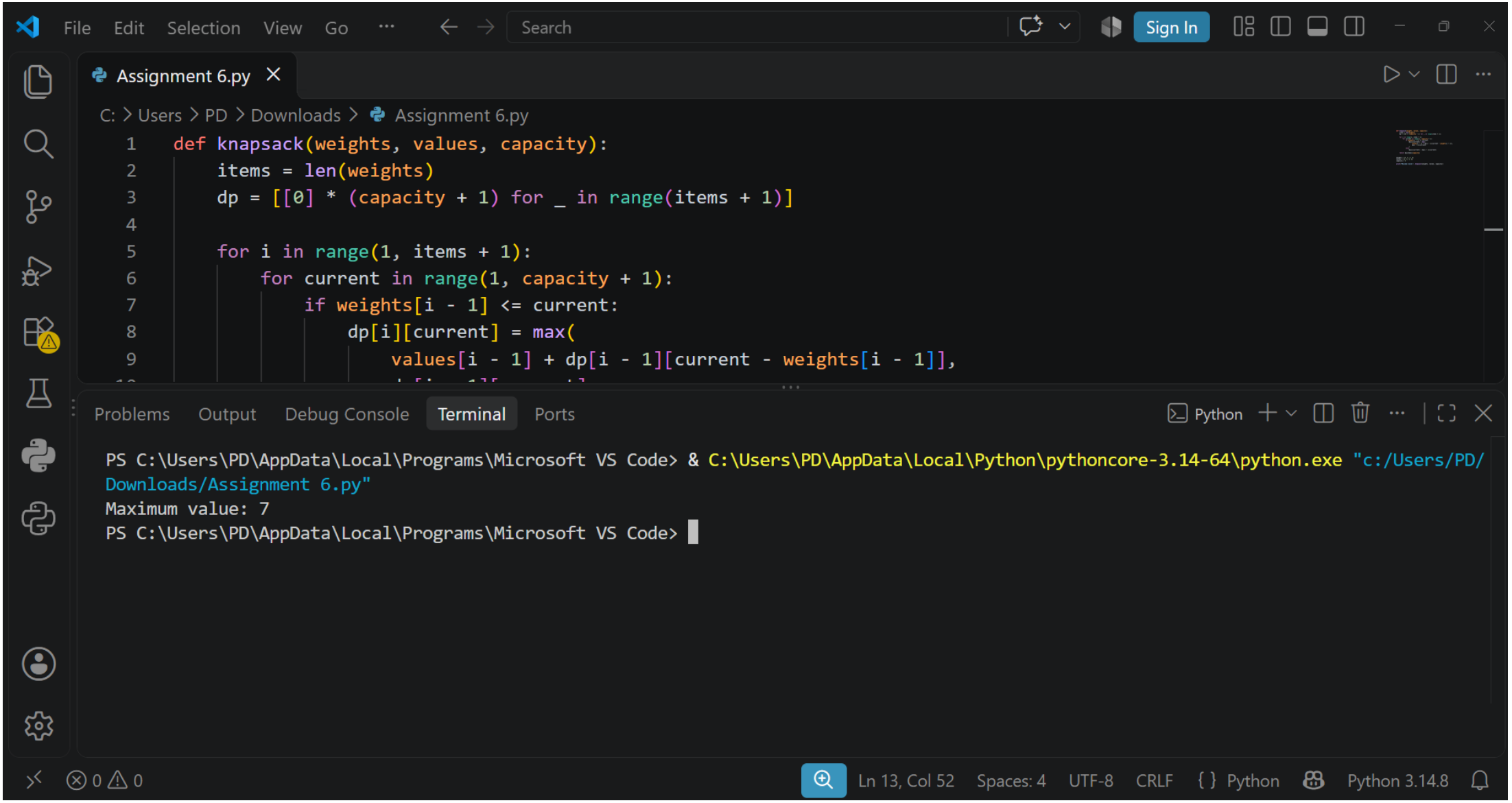

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Assignment 6.py
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def knapsack(weights, values, capacity):
items = len(weights)
dp = [[0] * (capacity + 1) for _ in range(items + 1)]
for i in range(1, items + 1):
for current in range(1, capacity + 1):
if weights[i - 1] <= current:
dp[i][current] = max(
values[i - 1] + dp[i - 1][current - weights[i - 1]],
dp[i - 1][current]
)
else:
dp[i][current] = dp[i - 1][current]
return dp[items][capacity]
weights = [2, 3, 4, 5]
values = [3, 4, 5, 6]
capacity = 5
print("Maximum value:", knapsack(weights, values, capacity))
0 0
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
{ } Python
Python 3.14.8


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Problems Output Debug Console Terminal Ports
Python
PS C:\Users\PD\AppData\Local\Programs\Microsoft VS Code> & C:\Users\PD\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/PD/Downloads/Assignment 6.py"
Maximum value: 7
PS C:\Users\PD\AppData\Local\Programs\Microsoft VS Code>
0 0
Ln 13, Col 52
Spaces: 4
UTF-8
CRLF
{ } Python
Python 3.14.8